Verilog

1 Complete the table with equivalents in Binary, Octal, and Hexa for : 143.75 (decimal), 11011.101 (binary), 453 (octal), 1F.C (Hexa)

Decimal	Binary	Octal	Hexadecimal
143.75			
	11011.101		
		453	
			1F.C

$143.75_{(10)} \rightarrow (?)_{(2)}$

2	143	
2	71	– 1
2	35	– 1
2	17	– 1
2	8	– 1
2	4	– 0
2	2	– 0
	1	– 0

= 10001111

$0.75 \times 2 = 1.5$ – 1

$0.5 \times 2 = 1$ – 1

$\therefore 143.75_{(10)} = 10001111.11_{(2)}$

$143.75_{(10)} \longrightarrow (?)_{(8)}$

8	143	
8	17	– 7
	2	– 1

= 217

$0.75 \times 8 = 6.0 = 6$

$\therefore 143.75_{(10)} = 217.6_{(8)}$

$143.75_{(10)} \rightarrow (?)_{(16)}$

$$\begin{array}{r|l} 16 & 143 \\ \hline & 8 - 15 \end{array} \uparrow = 8F \qquad 0.75 \times 16 = 12.0$$

$\therefore 143.75_{(10)} = 8F.C_{(16)}$

$11011.101_{(2)} \longrightarrow (?)_{10}$

$1\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0 + 1\times2^{-1} + 0\times2^{-2} + 1\times2^{-3} = 27.625_{(10)}$

$11011.101_{(2)} \rightarrow (?)_{(8)}$

$\underbrace{011}\,\underbrace{011}.\underbrace{101} = 33.5_{(8)}$

$11011.101_{(2)} \rightarrow (?)_{(16)}$

$\underbrace{0001}\underbrace{1011}.\underbrace{1010} = 1B.A_{(16)}$

$453_{(8)} \rightarrow (?)_{(10)}$

$4\times8^2 + 5\times8^1 + 3\times8^0 = 299_{(10)}$

$453_{(8)} \longrightarrow (?)_{(2)}$

$\underbrace{100}\,\underbrace{101}\,\underbrace{011} \longrightarrow 100101011_{(2)}$

$453_{(8)} \rightarrow (?)_{(16)}$

$\underbrace{0001}_{1}\underbrace{0010}_{2}\underbrace{1011}_{11} = 12B_{(16)}$

now $1F.C_{(16)} \rightarrow (?)_{(10)}$

$1\times16^1 + 15\times16^0 + 12\times16^{-1} = \underline{31.75_{(10)}}$

$1F.C_{(16)} \rightarrow (?)_{(2)}$

$\underbrace{0001}_{1}\underbrace{1111}_{15}.\underbrace{1100}_{12}{}_{(2)}$

$1F.C_{(16)} \rightarrow (?)_{(8)}$

$011\,111.110 \rightarrow \underline{37.6_{(8)}}$

2 Find Equivalents:

a) $324_{(5)} = (?)_{(10)}$

$3\times5^2 + 2\times5^1 + 4\times5^0 = \underline{89_{(10)}}$

b) $32.23_{(4)} = (?)_{10}$

$3\times4^1 + 2\times4^0 + 2\times4^{-1} + 3\times4^{-2} = \underline{14.6875_{(10)}}$

c) $231_{(10)} = (?)_{(7)}$

$$\begin{array}{r|l}
7 & 231 \\
7 & 33 - 0 \\
 & 4 - 5
\end{array}$$

$= \underline{450_{(7)}}$

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d) $189_{(10)} = (?)_{(6)}$

$$
\begin{array}{r|l}
6 & 189 \\
6 & 31 \;-\; 3 \\
 & 5 \;-\; 1
\end{array}
\qquad = 513_{(6)}
$$

3 Find the radix r:

a) $(312/20)_r = (13.1)_r$

$$\frac{312r}{20r} = 13.1r$$

$$\frac{3\times r^2 + 1\times r + 2}{2\times r + 0} = 1\times r^1 + 3\times r^0 + 1\times r^{-1}$$

$$\frac{3r^2 + r + 2}{2r} = r + 3 + \frac{1}{r}$$

$$3r^2 + r + 2 = 2r^2 + 6r + 2$$

$$r^2 - 5r = 0$$

$$r^2 = 5r$$

$$\boxed{r = 5}$$

b) $264r + 136r = 433r$

$$2\times r^2 + 6\times r^1 + 4\times r^0 + 1r^2 + 3\times r^1 + 6\times r^0 = 4\times r^2 + 3\times r^1 + 3\times r^0$$

$$2r^2 + 6r + 4 + r^2 + 3r + 6 = 4r^2 + 3r + 3$$

$$3r^2 + 9r + 10 = 4r^2 + 3r + 3$$

$$-r^2 + 6r + 7 = 0$$

$$r = -1 \qquad r = 7$$

r cannot be negative

$\therefore \boxed{r = 7}$

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c) $221505_{(8)} = 12345r$

$2\times8^5 + 2\times8^4 + 1\times8^3 + 5\times8^2 + 0\cancel{*}5\times1 = 1\times r^4 + 2r^3 + 3r^2 + 4r + 5$

$r = 16$

4 Addition

a) $1101_{(2)} + 11011111_{(2)}$

$$\begin{array}{r} 1101 \\ +\ 11011111 \\ \hline 11101100 \end{array}$$

$= 11101100_{(2)}$

b) $77652_{(8)} + 76543_{(8)}$

$$\begin{array}{r} 77652 \\ +\ 76543 \\ \hline 1764\cancel{8}15 \end{array}$$

$5+4 = 9_{(10)} = 8\underline{|9}$ $1-1$ $=11$

$6+6 = 12_{(10)}$ $8\underline{|12}$ $1-4$ $=14$

$= 176415_{(8)}$

$7+7 = 14_{(10)}$ $8\underline{|14}$ $1-6$ $=16$

$8+7 = 15_{(10)}$ $8\underline{|15}$ $1-7$ $=17$

c) $578A_{(16)} + AB\cancel{C}DE_{(16)}$

$$\begin{array}{r} 578A \\ +\ ABDE \\ \hline 10368 \end{array}$$

$A+E = 24_{(10)}$ $16\underline{|24}$ $1-8$ $=18_{(16)}$

$9+D = 22_{(10)} = 16\underline{|22}$ $1-6$ $=16_{(16)}$

$8+B = 19_{(10)} = 16\underline{|19}$ $1-3$ $=13_{(16)}$

$10368_{(16)}$

$6+A = 16_{(10)}$ $16\underline{|16}$ $1-0$ $=10_{(16)}$

d) $234_{(6)} + 315_{(6)}$

$$\begin{array}{r} 234 \\ +\ 315 \\ \hline 553 \end{array}$$

$5+4=9_{(10)}$ $\quad 6\,|\,9$ $\quad 1-3 \quad = 13_{(6)}$

$3+2=5_{(10)}$ $\quad 6\,|\,5$ $\quad 0-5 \quad = 05$

$553_{(6)}$

5 Binary → Gray code Convertion
(xor operation / bitwise xor)

a) $10101_{(2)}$

$1 \xrightarrow[xor]{} 0 \xrightarrow[xor]{} 1 \xrightarrow[xor]{} 0 \xrightarrow[xor]{} 1_{(2)}$

$= 1111_{(2)}$

b) $01110_{(2)}$

$0 \xrightarrow[xor]{} 1 \xrightarrow[xor]{} 1 \xrightarrow[xor]{} 1 \xrightarrow[xor]{} 0_{(2)}$

$= 01001_{(2)}$

6 Gray code → Binary

a) 01101 (gray)

0 1 1 0 1 (gray)

xor xor xor xor

$0\ 1\ 0\ 0\ 1_{(2)}$ $\qquad 01001_{(2)}$

b) 10101 (gray)

1 0 1 0 1 (gray)

xor xor xor xor

$1\ 1\ 0\ 0\ 1_{(2)}$ $\qquad 11001_{(2)}$